Draw = dose ÷ concentration. IV drugs need an IV or IO first. More than the range adds side effects, not benefit.

FIXED DOSES

DRUG	FOR	DOSE	DRAW
Epinephrine	Cardiac arrest	1 mg IV	1 ml
Amiodarone	VT or VF	300 mg IV	6 ml, two vials
Lidocaine	VT or VF	100 mg IV	5 ml
Lidocaine	Before realignment or thoracostomy	50 to 100 mg IM	2.5 to 5 ml
TXA	Severe bleeding, bruising, bleeding hemothorax	1 g IV, up to 2 g	10 ml
Calcium chloride	After the first unit of blood, then every 4 units	1 g IV, never IM	10 ml
Ondansetron	Nausea, vomiting, after ketamine	4 mg IV or IM, max 8 mg	2 ml
Ertapenem	Wound infection, before conversion if required	1 g IV or IM	3.2 ml
Atropine	Nerve agent	2 mg IV or IM, 2 to 4 mg to start	2 ml, two vials
Atropine	Slow heart rate after amiodarone	1 mg IV or IM	1 ml
Adenosine	Lowers heart rate	6 mg IV	2 ml

WEIGHT DOSES, WORKED FOR 80 KG

DRUG	ROUTE	PER KG	AT 80 KG	DRAW
Morphine	IV	0.05 to 0.1 mg	4 to 8 mg	0.8 to 1.6 ml
Morphine	IM	0.07 to 0.1 mg	5.6 to 8 mg	1.1 to 1.6 ml
Fentanyl	IV	0.5 to 1 mcg	40 to 80 mcg	0.8 to 1.6 ml
Fentanyl	IM	0.7 to 1 mcg	56 to 80 mcg	1.1 to 1.6 ml
Ketamine	IV	0.1 to 0.2 mg	8 to 16 mg	0.16 to 0.32 ml
Ketamine	IM	0.4 to 0.8 mg	32 to 64 mg	0.64 to 1.28 ml
Lidocaine	IV	1 mg, max 3 mg	80 mg, max 240 mg	4 ml, max 12 ml
Esmolol	IV	0.25 to 0.5 mg	20 to 40 mg	2 to 4 ml

TIMINGS AND VITALS

DRUG	ONSET IV / IM	LASTS	VITALS
Morphine	Under 25 s / 2 min	21 / 30 min	Lowers HR, BP, breathing
Fentanyl	Under 10 s / 30 s	16 / 20 min	Lowers HR, BP, breathing
Ketamine	Under 5 s / 20 s	11 / 15 min	None
Fentanyl lozenge	Under 2 min	15 min	Lowers breathing most
Epinephrine	Under 5 s / 30 s	6 to 7 min	Raises HR, BP, breathing
Amiodarone	Under 10 s	12 min	Lowers HR, BP
Lidocaine	Under 5 s	10 min	Lowers HR, BP
Atropine	Under 5 s / 15 s	15 min	Raises HR
Esmolol	Under 20 s	5 min	Lowers HR
Adenosine	Under 1 min	Under 2 min	Lowers HR
TXA	Under 15 s	15 min	Lowers BP slightly
Calcium chloride	Under 15 s	10 min	
Ondansetron	Under 15 s / 45 s	12 to 15 min	
Ertapenem	Under 5 s / 30 s	15 to 20 min	

DO NOT MIX

- No amiodarone and lidocaine within 5 to 10 minutes of each other. No second amiodarone within 5 to 10 minutes.
- No opioids with low blood volume, slow heart rate, slow breathing, chest injury or altered mental status.
- Naloxone for any arrest with opioids on board.
- Calcium chloride never IM, and not in ventricular fibrillation.
- No epinephrine with a very fast heart rate or high blood pressure.
- No esmolol, adenosine, amiodarone or midazolam with a slow heart rate.
- Morphine autoinjector is not issued.

PAIN: WHICH DRUG

SITUATION	DRUG
Breathing impaired or chest injury	Ketamine. Consider ondansetron 4 mg after
No breathing problems	Morphine. Fentanyl if severe
Conscious, lying down, no IV	Fentanyl lozenge, 1